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PII: S1572-6657(19)30083-9
DOI: https://doi.org/10.1016/j.jelechem.2019.01.069
Reference: JEAC 12899

To appear in: *Journal of Electroanalytical Chemistry*

Received date: 9 December 2018
Revised date: 26 January 2019
Accepted date: 29 January 2019

Please cite this article as: S. Schindler and T. Bechtold, Mechanistic insights into the electrochemical oxidation of dopamine by cyclic voltammetry, Journal of Electroanalytical Chemistry, https://doi.org/10.1016/j.jelechem.2019.01.069

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Mechanistic insights into the electrochemical oxidation of dopamine by cyclic voltammetry

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Abstract

Due to its presence in many biological systems, the oxidation of dopamine to polydopamine is of high interest with regard to biomedical and technical applications. Many variations in the reaction mechanisms have been discussed in the scientific literature, indicating a strong influence of the particular reaction conditions chosen. In this paper a cyclic voltammetric study on a gold electrode in phosphate buffered saline at pH 5.8 and 7.0 has been performed to elucidate the anodic oxidation at physiologic pH conditions. Calculation of the pH dependent species distribution of dopamine was used to identify the relevant species involved in the first step of the anodic oxidation. In the pH range 5.8 – 7.0 the phenolic groups of dopamine are not present in dissociated form and also the amino group is present in protonated form. The oxidation at the gold electrode is an electrochemically reversible process and leads to formation of a semiquinone as an intermediate. The follow-up reactions of the anodically formed intermediates depend on the solution pH. At pH 7.0 the formation of brown polymeric products on the electrode is faster and compared to experiments at pH 5.8 a lower cathodic peak current $(I_p)_c$ is observed during the reverse scan. The semiquinone intermediate explains formation of ring-bridged dimers as oligomeric precursors in polydopamine formation. Even at pH 7 the amino group still is present as ammonium ion,

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thus Michael addition, Schiff type reactions and cyclisation to indole ring containing products are expected to proceed as follow-up reactions with lower reaction rate.

Keywords: cyclic voltammetry; dopamine; polydopamine; oxidation; coating

1. Introduction

The catecholamine dopamine is a hormone essential for our well-being (Figure 1). Dopamine is a rather reactive molecule that oxidizes easily to polydopamine at physiological conditions. Polydopamine is a melanin-like [1], black polymeric aggregate. Polydopamine has been demonstrated to show protective effects on human cells, at the same time highly reactive intermediate species are formed, which provoke severe damage through oxidative stress. Both effects, of protective and destructive kind, are observed in Parkinson's Disease [2]. But not only inside, also outside of our bodies dopamine has gained much interest throughout the last decades: it was in 1981 when L- DOPA has been discovered as the "glue" for mussels [3] to attach efficiently to wet surfaces. This discovery has inspired scientists to make use of the unique molecule dopamine. In material sciences, Lee et al. have developed the first oxidative polydopamine coating-method [4] in 2007 by auto-oxidation of dopamine in slightly alkaline medium.

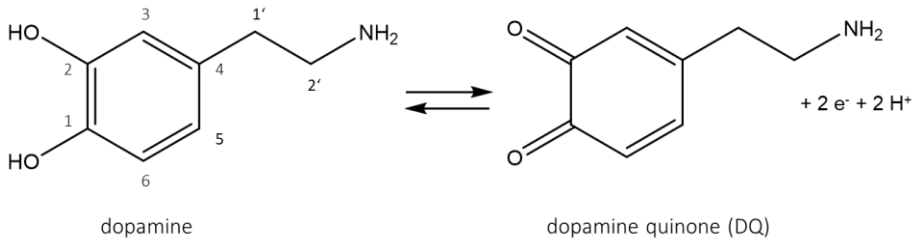


Figure 1: Structure of 4-(2'-aminoethyl)benzene-1,2-diol (IUPAC name for dopamine) and oxidation to Dopamine Quinone (DQ) in a 2-electron oxidation.

There are four main ways to stimulate polydopamine-formation for coating: by (1) alkaline auto-oxidation [4,5] or (2) using strong oxidizing agents [6], by (3) UV-triggered oxidation [7,8] or (4) electrochemically-induced oxidation [9]. In however way polydopamine-formation is induced, the oxidation pathway of dopamine is a complex multi-step reaction which involves several steps and reaction intermediates of dopamine. The first oxidation product of dopamine is dopamine quinone (DQ, see Figure 1), a highly reactive intermediate and the starting point for polydopamine formation.

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It was already in 1967 when the electrochemical behaviour and oxidation mechanism of catechols [10] have been studied in detail. Soon the "electrode fouling" of catecholamines has become a well-known phenomenon [11]. A detailed study on electrochemical deposition of dopamine has shed new light on how the coating-ability of dopamine depends on pH, dopamine concentration and ionic strength [9]. The use of oxidizing agents like copper [12] or $NaIO_4$ [6] speeds up the overall reaction and enables coating also in acidic environments. Also UV-triggered oxidation [7,8] has been employed like a switch to turn polydopamine formation "on", while antioxidants like ascorbic acid may turn the reaction "off".

The oxidative formation of polymer layers based on dopamine are of high interest for antibacterial [13,14] and antifouling coatings [15,16]. In solutions of pH = 7 and higher, the oxidation of dopamine to the brown-black coloured polydopamine proceeds spontaneously through action of the dissolved oxygen. The spontaneous oxidation of dopamine can be used to coat a metal surface e.g. by dipping the substrate into a solution containing dopamine at pH 8.5 or higher. The polydopamine-layer will form through oxidation of dopamine by O_2, which however is a time-consuming procedure and hard to control.

For the formation of polydopamine layers as surface coatings on metal surfaces anodic oxidation of dopamine could be a preferable approach to concentrate the oxidative reaction near to the surface of the metal anode. Coating via electrochemically-induced oxidation allows for uniform coatings in short times, just a few seconds - with a minimal waste of material. As dopamine is with 3.2 USD per gram a rather expensive chemical, this approach is of high relevance for larger scale applications [17].

Li and co-workers [9] investigated the electrochemical deposition of dopamine by use of a quartz crystal microbalance to study the polydopamine growth on Au surfaces.

They investigated dopamine in phosphate buffered saline (PBS) with different dopamine concentrations at distinct pH values between 2 and 11. For this setup, an ECE mechanism has been proposed, which at least involved transfer of 4 electrons and release of 4 H^+, 2 each per E step. Another earlier study on electrochemical oxidation of dopamine at physiological pH proposes an ECC in a sequence of transfer of 2 electrons and release of 2 protons (E), ammonium deprotonation (C) and Michael-type ring closure (C) [18].

In the cyclic voltammetry of quinones the mechanism of the cathodic reduction of a quinone is separated into a series of EC reactions which involve 1 e^- and 1 H^+ [19]. In analogy in the anodic oxidation of dopamine, radicals could be involved as intermediates. However this very first step has not been considered in sufficient depth for the anodic oxidation mechanism of dopamine.

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In this study, the critical pH-window between pH 5 and 7, where auto-oxidation begins to occur, has been chosen for a detailed study of the anodic oxidation of dopamine on solid-state electrodes. Calculations of the pH-dependent species distribution of dopamine were used to identify the major species involved in the initial anodic oxidation. Cyclic voltammetry experiments at different scan rates were used to elucidate the mechanism of the anodic dopamine oxidation in detail and to investigate for a possible radical-based reaction mechanism.

2. Material & methods

2.1 Chemicals and reagents

The chemicals used were analytical grade quality:

Dopamine hydrochloride (Sigma Aldrich, Steinheim, Germany), NaCl (≥99.5%, Carl Roth, Karlsruhe, Germany), $K_4[Fe(II)(CN)_6]x3H_2O$ (potassium hexacyanoferrate(II)trihydrate, puriss. ≥99%, Sigma Aldrich, Steinheim, Germany), $K_3[Fe(III)(CN)_6]$ (potassium hexacyanoferrat(III), p.a. ≥99%, Fluka AG, Seelze, Germany), KH_2PO_4 (potassium dihydrogen phosphate, p.a. ≥98.5%, E. Merck, Darmstadt, Germany), $Na_2HPO_4x2H_2O$ (disodium hydrogen phosphate dihydrate, p.a. ≥99.5%, E. Merck, Darmstadt, Germany), acetic acid (96%, Carl Roth, Karlsruhe, Germany), sodium acetate (anhydrous, Fluka AG, Seelze, Germany).

0.1 M phosphate buffer was prepared from 0.1 M solutions of KH_2PO_4 and Na_2HPO_4 by mixing to give the desired pH of 5.8 or 7.0. pH was measured using a Mettler Toledo MP225 pH meter (Mettler Toledo, Vienna, Austria). For phosphate-buffered saline (PBS), 0.5 M NaCl was added to the phosphate buffer. 0.05 M acetate buffer was prepared in equal manner from 0.05 M solutions of sodium acetate and 0.05 M acetic acid. Pure solutions were mixed under pH control until pH set to 5.0. Buffers were stored at 4°C in the dark. Solutions of 0.5, 2 and 5 mM dopamine hydrochloride were prepared only prior to use as dopamine oxidizes quickly in solution. 5 mM dopamine hydrochloride corresponds to 1 mg/ml.

2.2 Cyclic voltammetry

Cyclic voltammetry was carried out employing a three-electrode configuration. The apparatus used for the experiments was an EG&G 264 A potentiostat using a gold working electrode, 2 mm in diameter (3.14 mm^2). A Platinum wire served as counter electrode. The cyclic voltammograms were recorded on a Rikadenki X–Y recorder. All potentials given were measured against an Ag/AgCl (3M KCl) reference electrode. A redox buffer according DIN

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38404 part 6 [20] was used to inspect the reference electrode. The buffer was prepared through dissolution of 0.528 g potassium hexacyanoferrate(II)trihydrate ($K_4[Fe(CN)_6]x3H_2O$), 0.411 g potassium hexacyanoferrate(III) ($K_3[Fe(CN)_6]$), 0.180 g potassium dihydrogenphosphate (KH_2PO_4) and 0.390 g di-sodium hydrogenphosphate dihydrate, ($Na_2HPO_4x2H_2O$) in 100 ml deionised water and was stored in the dark. Before measuring, all test solutions were de-oxygenated for 8 min with Ar (inert gas) to eliminate interfering oxygen. All experiments were performed at room temperature.

Before measuring, the potential of the reference electrode was checked using the redox buffer. Then, a first dataset was recorded: three different dopamine concentrations (0.5, 2 and 5 mM) in 0.1 M phosphate buffer at two different pH values (5.8 and 7.0) and five scan rates (5, 10, 20, 50 and 100 mV/s) were investigated by cyclic voltammetry. After each cyclic voltammetry experiment, the electrode was cleaned with abrasive Zeolith powder (Wessalith P, Degussa, Frankfurt, Germany) and rinsed with water. Having finished the measurements, the potential of the reference electrode was checked again. All experiments were recorded in duplicate. Cyclic voltammograms were evaluated and means of duplicates are presented.

3. pH dependent species distribution of dopamine

Dopamine contains one primary amino function that can be protonated and two phenolic groups that can dissociate. Thus, three pKa values are assigned to dopamine, which give rise to three dissociation equilibria [21]. Species distribution (α_i) of dopamine as function of pH is shown in Figure 2. While there have been plenty of studies on the deprotonation order of dopamine's functional groups published in the literature, there is still discussion ongoing about the site of first dissociation in **I**-$DPAH_3^+$, in particular the position of the first phenolic hydroxyl group to dissociate.

In literature, different values for the pKa of dopamine have been reported, dependent on the experimental conditions used for determination e.g. temperature, ground electrolyte. The presentation of a species distribution as function of pH given in Figure 2 bases on the values of Sanchez-Rivera et al. [21] which have been determined by spectrophotometric titration at 20 °C and a dopamine concentration of 0.125 mM and ionic strength of 0.5 M NaCl in combination with point-by-point analysis. Species distribution (α_i) has been calculated by CurTiPot [24] using the aforementioned pKa values. The pH values used for the CV experiments in this study are indicated by the red dashed lines. As can be seen, at pH 5.8 as well as at pH 7 the fully protonated species **I**-$DPAH_3^+$ will dominate.

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Complexation experiments of several catechols with Al^{3+}, Kiss, Sovago and Martin [22] verified that the zwitterionic species **II** shown in Figure 2 is 12-times more abundant than the neutral species that results from ammonium instead of phenol deprotonation.

In this work we use a deprotonation scheme with species order **I-IV** which is in accordance with other groups [23]. Molecular structures of species **I-IV** are shown in detail in Figure 2. The depicted deprotonation scheme is also in accordance with the values pKa_1 (phenol) and pKa_2 ($-NH_3^+$).

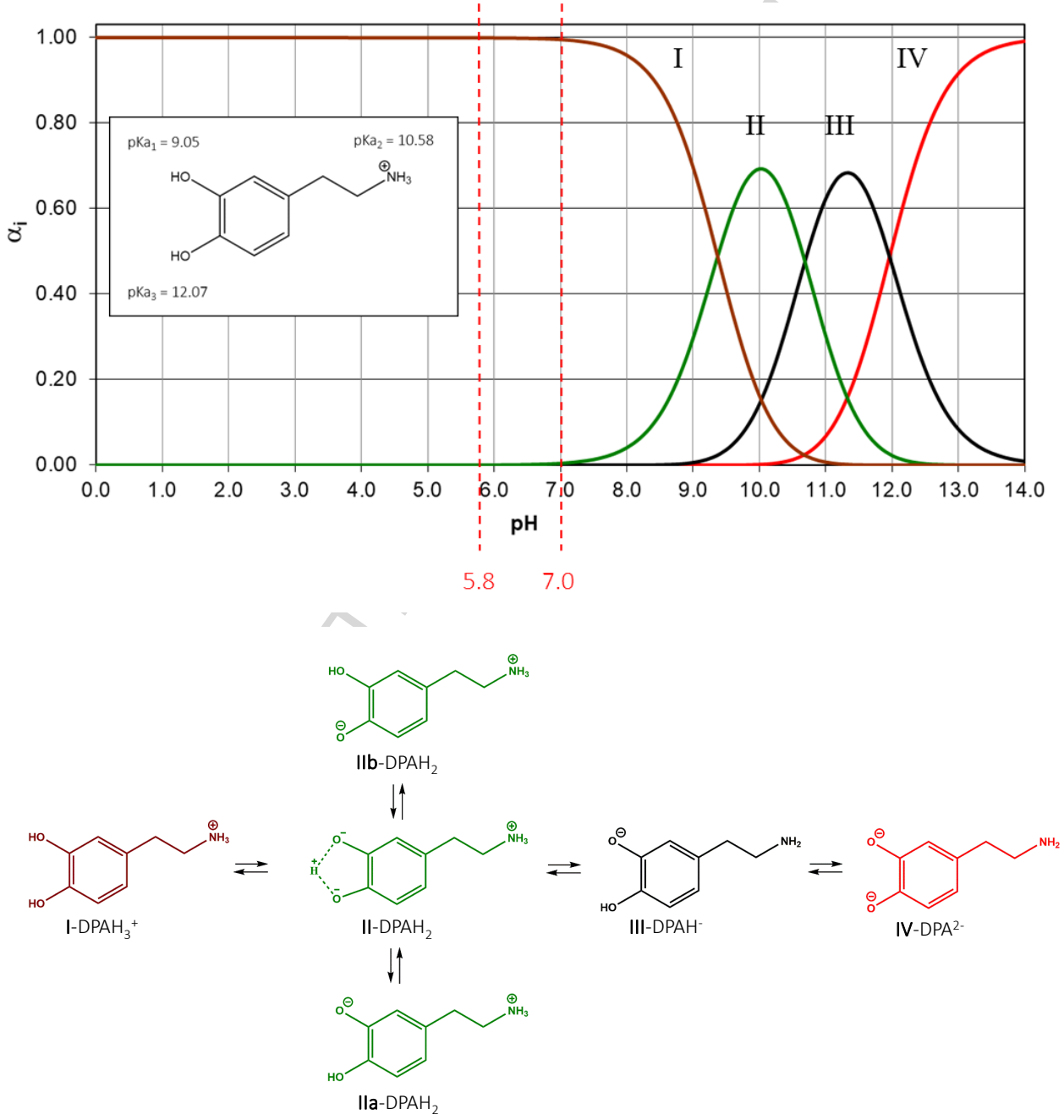


Figure 2: I-$DPAH_3^+$, **II/IIa/IIb**-$DPAH_2$, **III**-$DPAH^-$, **IV**-DPA^{2-}. Species distribution of DPA in aqueous solution as function of pH using pKa values as determined by Sanchez-Rivera et al. [21]. Thereof, species distribution has been calculated by CurTiPot [24].

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Corona-Avendano et al. investigated ^{13}C-NMR shifts [25] for dopamine at pH 5.1 vs. pH 9.6. At pH 5.1 the fully protonated species I-DPAH^{3+} predominates, while at pH 9.6 the second species **II**-DPAH$_2$ is predominant. The NMR results interestingly indicate a first dissociation at the catechol site (C1, C2 in Figure 1) for pH 9.6, the shift for the carbon next to the ammonium function (C1') remains unaffected.

Addressing the second question, which of the phenolic groups will be deprotonated preferentially as first step, it is very likely that the negative charge arising after the first phenolic deprotonation will be shared between both phenolic oxygen atoms. Thus, an intramolecular hydrogen bonding will stabilize the remaining proton (**II**-DPAH$_2$).

Taking the published analytical and theoretical information about dissociation of dopamine into account, we propose the deprotonation scheme shown in Figure 2. This scheme is also supported by the aforementioned ^{13}C-NMR shifts that have been recorded by Corona-Avendano et al. where both phenolic carbon atoms do show the same alteration of chemical shifts for pH 9.6: a Δ of 3.81 and 3.61 ppm respectively [25]. This indicates that the electronic surroundings of both phenolic carbons are equal.

For the formulation of an anodic dopamine oxidation mechanism under the conditions applied in this work thus we will have to consider species **I**-DPAH$_3^+$ as starting point.

4. Results and Discussion

4.1 Electrochemical analysis of anodic oxidation of dopamine

Different dopamine concentrations and scan rates were reported in the literature for electrochemical experiments. Referring to Li et al. 5 mM dopamine was chosen as the optimal deposition concentration, as with lower concentrations no formation of polydopamine had been achieved [9].

Cyclovoltammograms of 0.5, 2 and 5 mM dopamine were recorded in 0.1 M phosphate buffer at pH 5.8 and 7.0 at scan rates of 5, 10, 20, 50 and 100 mV/s to analyse the electrode reactions at different time scale. In Figure 3 representative cyclovoltammograms recorded with 5 mM dopamine hydrochloride in 0.1 M phosphate buffer at pH 5.8 and pH 7.0 are shown.

From auto-oxidation experiments in presence of air oxygen in the pH range 5.8 – 7.0 (Figure 4) it could be shown, that formation of coloured polymeric oxidation products rapidly occurs at pH 7.0, while in solutions of lower pH dopamine oxidation is substantially lower. Thus buffered dopamine solutions at pH 5.8 and pH 7.0 were chosen for the experiments.[9] All six

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sets of cyclovoltammograms were evaluated for their anodic and cathodic peak potentials $(E_p)_a$ and $(E_p)_c$ and the corresponding peak currents $(I_p)_a$ and $(I_p)_c$ (Table 1).

Table 1. Cyclic voltammograms of dopamine solutions at pH 5.8 and pH 7.0 on an Au-electrode (3.14 mm²). Anodic (oxidation) and cathodic (reduction) peak potentials $(E_p)_a$ and $(E_p)_c$ and the corresponding peak currents $(I_p)_a$ and $(I_p)_c$ as function of concentration of dopamine and scan rate.

		pH 5.8					**pH 7.0**				
c(dopamine) mM	scan rate *v* mV/s	$(E_p)_a$ mV	$(I_p)_a$ µA	$(E_p)_c$ mV	$(I_p)_c$ µA	$(I_p)_c/(I_p)_a$	$(E_{p)a}$ mV	$(I_p)_a$ µA	$(E_p)_c$ mV	$(I_p)_c$ µA	$(I_p)_c/(I_p)_a$
	5	510	-1.9								
	10	533	-2.8	120	0.6	0.23	405	-2.8			
0.5	20	555	-4.1	120	1.1	0.27	420	-3.8			
	50	618	-6.5	100	2.4	0.37	445	-5.5	63	0.5	0.09
	100	670	-8.9	105	4.1	0.46	465	-7.4	53	1.8	0.24
	5	488	-6.3	123	1.0	0.16	455	-6.0			
	10	495	-7.3	118	2.8	0.38	560	-7.6			
2.0	20	545	-11.8	108	4.8	0.40	580	-11.1			
	50	600	-19.0	85	8.8	0.46	515	-16.6	30	1.6	0.10
	100	615	-26.3	80	13.3	0.50	535	-24.4	30	5.9	0.24
	5	495	-12.5	118	2.0	0.16	403	-14.8			
	10	540	-17.5	110	6.5	0.37	430	-20.5			
5.0	20	600	-25.5	100	13.0	0.51	400	-31.3			
	50	620	-41.0	98	23.0	0.56	405	-48.0	38	3.0	0.06
	100	635	-59.0	93	31.5	0.53	440	-66.0	38	9.8	0.15

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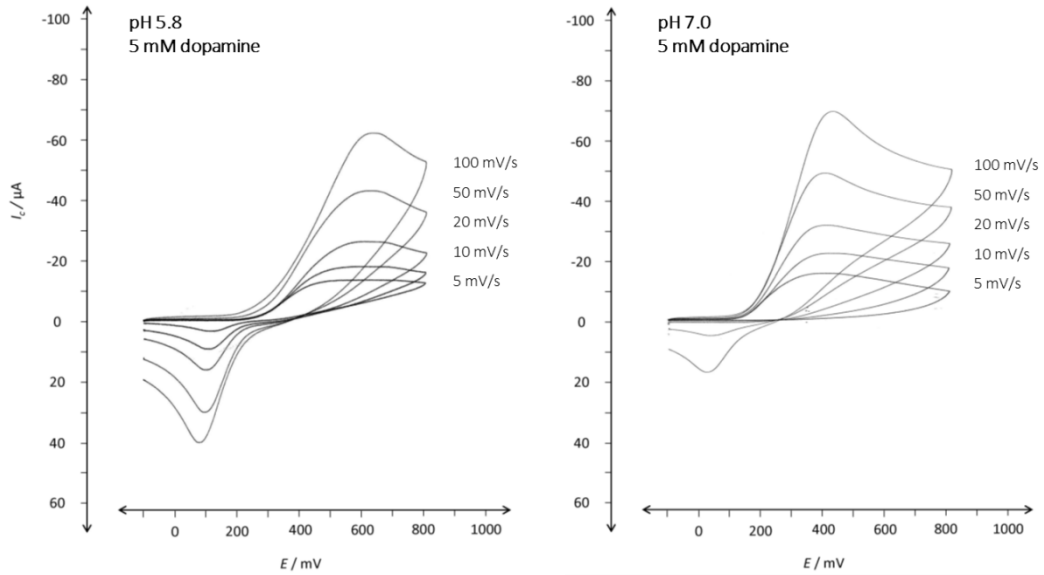


Figure 3: Cyclic voltammograms of 5 mM dopamine solutions at pH 5.8 and pH 7.0 in the potential interval of -100 - +800 mV recorded on an Au-electrode (3.14 mm²) at different scan rates (5, 10, 20, 50, 100 mV/s).

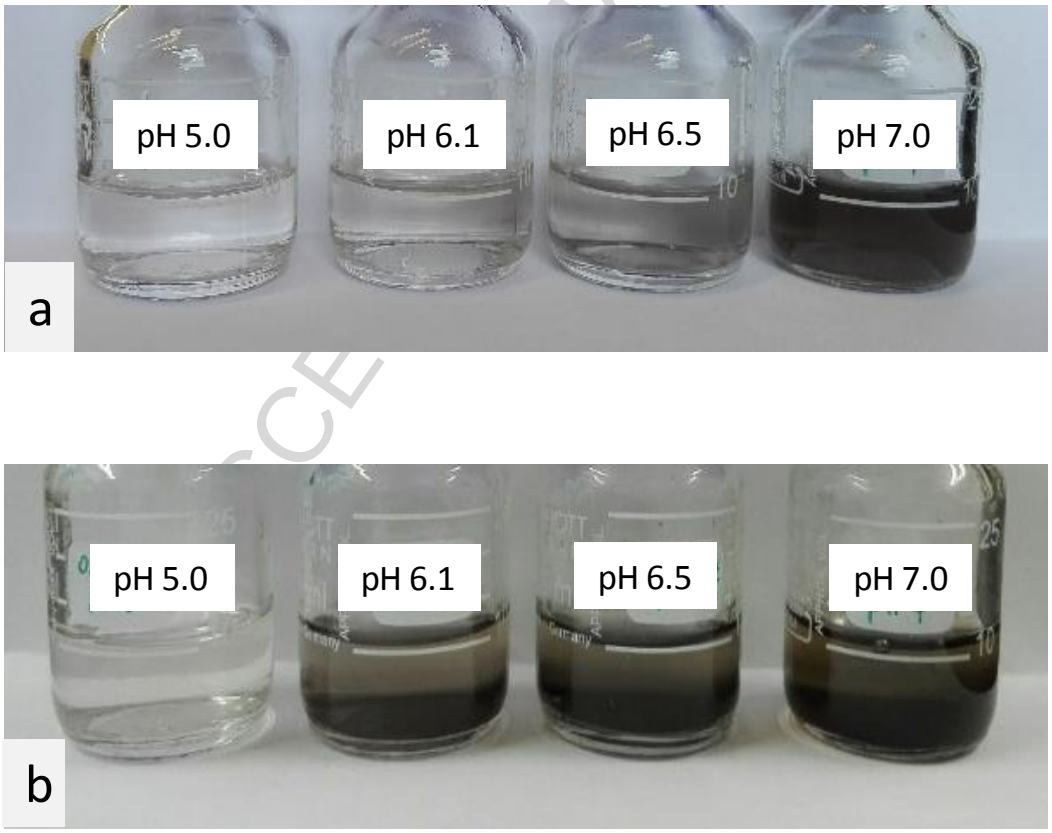


Figure 4: pH-dependent auto-oxidation of dopamine. Order from left to right: pH 5, pH 6.1, pH 6.5, pH 7 **a)** after 24 h of oxidation, **b)** after 96 h of oxidation at RT.

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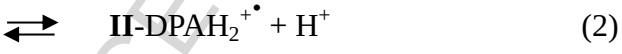
Figure 4 shows time dependent formation of coloured products for pH 5 and 7. At pH 8 or above the oxidation reaction is fast and dark aggregates form quickly, while at pH 5 even after 4 days at room temperature hardly any aggregates were visible, which is in agreement to literature [5].

Comparing peak potentials $(E_p)_a$ and $(E_p)_c$ for pH 5.8 and pH 7.0, we notice a shift towards lower potentials for experiments at pH 7 both for the anodic oxidation and the cathodic reduction peak. The shift in potential indicates an involvement of protons in the overall redox reaction, as expected from mechanisms already proposed in the literature [9,18,26,27].

At lower dopamine concentration the anodic peaks at pH 7.0 are smaller compared to pH 5.8, while at 5 mM dopamine anodic peak current values $(I_p)_a$ is approximately 20 % higher for pH 7.0 than for pH 5.8. At a dopamine concentration of 5 mM and a scanning rate of 50 mV/s, the anodic peak increases from -41 µA at pH 5.8 to -48 µA at pH 7.0. The minor differences in peak current between the two pH values investigated are in agreement with the proposed species distribution for dopamine. In the pH interval of pH 5.8 – 7.0 the fully protonated species **I**-$DPAH_3^+$ has been predicted as the major species present in the diffusion layer and being thus available for anodic oxidation. As a result, no distinct change in anodic peak currents is expected. As a consequence, we can assume that the same product is formed as first step of the anodic oxidation (equation 1).



I-$DPAH_3^+$ ⇌ **I**-$DPAH_3^{2+\bullet}$ + e^- (1)



I-$DPAH_3^{2+\bullet}$ ⇌ **II**-$DPAH_2^{+\bullet}$ + H^+ (2)

In agreement to the literature results published for the cyclic voltammetry of anthraquinones dissociation of the initially formed radical can be formulated as second step (equation 2) [19].

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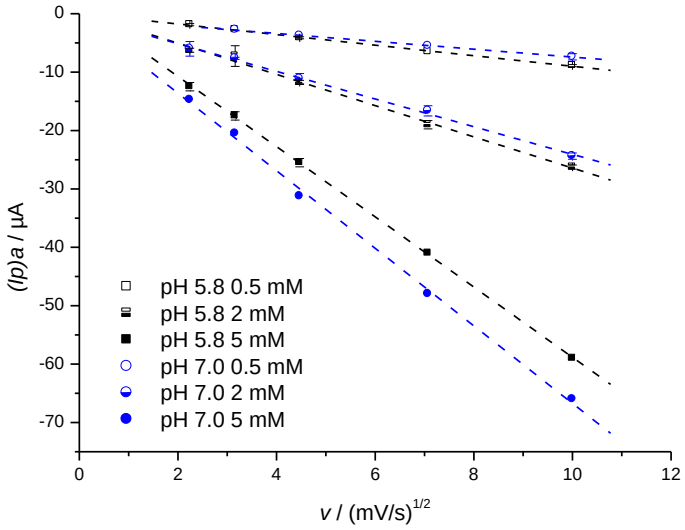


Figure 5: Anodic peak current $(I_p)_a$ as function of square root of scan rate $v^{1/2}$ for the oxidation of dopamine at pH 5.8 and pH 7 in the potential interval of -100 - +800 mV recorded on a Au-electrode (3.14 mm²) at different scan rates (5, 10, 20, 50, 100 mV/s) (mean of two repetitions and standard deviation as error bars).

An analysis of the electrochemical reversibility is shown in Figure 5 where the anodic peak current $(I_p)_a$ has been drawn as function of square root of scan rate ($v^{1/2}$). A linear relationship is observed independently of pH value used, which demonstrates an electrochemically reversible electron transfer reaction.

4.2 Oxidation of dopamine

Dopamine is readily oxidized to form melanin-like aggregates. Often the oxidation mechanism to polydopamine is formulated to start from the neutral structure of dopamine (Figure 1). However, the protonation state at a given pH will be of strong influence to the reaction mechanism and so also the oxidation mechanism to polydopamine will strongly depend on pH. At acidic and neutral pH, only two out of the four possible species will be present in solution (Figure 2). Depending on pH and protonation state of the amino and phenol functions, different reaction pathways to polydopamine will predominate (Figure 6).

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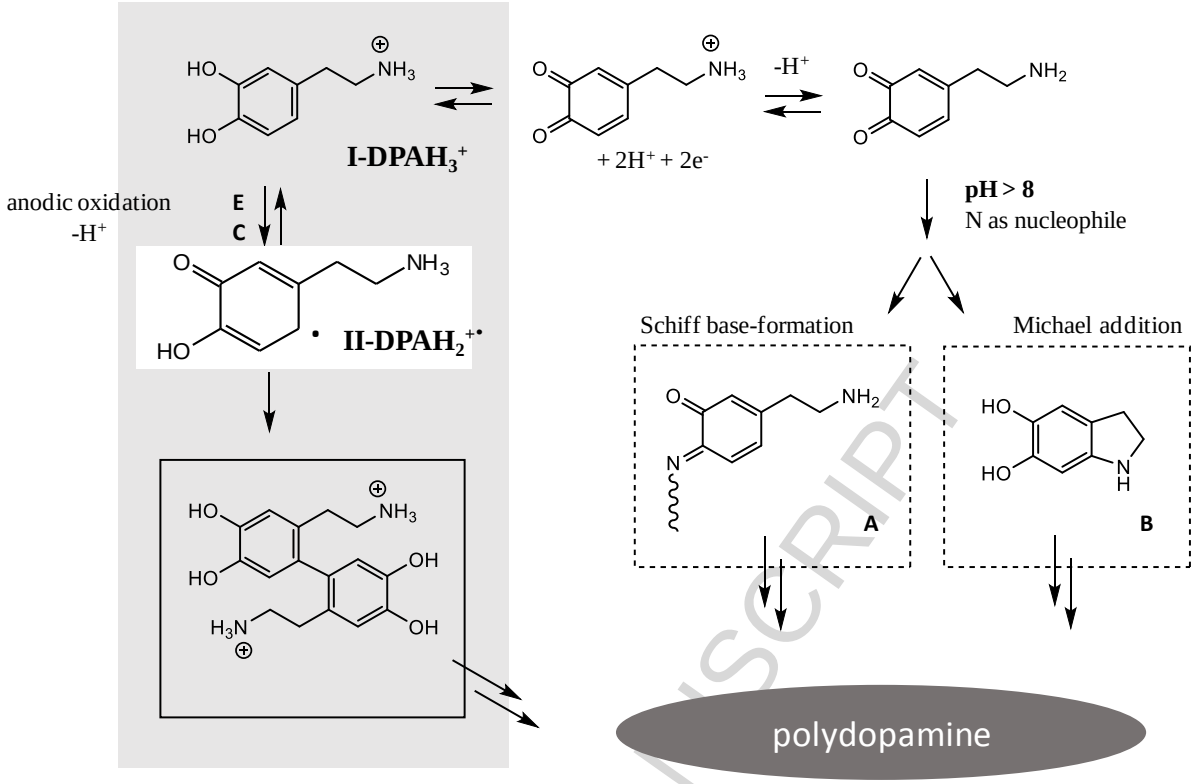


Figure 6: pH dependent reactions pathways of dopamine leading to formation of polydopamine. Starting point for the oxidation at pH 5.8 and pH 7.0 will be the protonated species **I**-$DPAH_3^+$. The oxidation main reaction route is marked in grey.

A general reaction scheme for polydopamine formation based on literature results is summarised in Figure 6 and has been condensed from different literature sources [9,18,26,27]. Out of the methods of polydopamine formation proposed therein we hypothesise that the anodic oxidation will include the transfer of single electrons and thus radical formation is expected as first step of the reaction pathway. Semiquinone formation has already been proposed as intermediate product present during oxidation of dopamine with sodium periodate [28]. The proposed reaction mechanism for anodic oxidation of dopamine is highlighted in gray and involves formation of a semiquinone radical after a one-electron transfer reaction (E) at the anode with subsequent deprotonation (C). The semiquinone radical then dimerises, if the dopamine/radical concentration is high enough. The reaction mechanisms instead that are commonly proposed in literature are shown as routes **A** and **B** and involve reaction of the amine group as a nucleophile. According to the pKa value of the amino function (pKa_2 = 10.58 [21]) this group will be available in non-protonated form only at higher pH values. At pH 8.5 only 0.1% of all dopamine molecules will contain the non-protonated amino group.

4.3 Reaction mechanism for anodic formation of polydopamine

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In the CV experiments the initially formed **II**-$DPAH_2^{+\bullet}$ will undergo follow up reactions to form coloured products. While the anodic peak current exhibits low dependency on the pH of the buffered solution, the corresponding cathodic current $(I_p)_c$ for the reduction of oxidation products formed exhibits a distinct decrease with change of solution pH from 5.8 to 7.0. As an example, cathodic peak current $(I_p)_c$ measured with a pH 5.8 in a 5 mM dopamine solution at 50 mV/s decreased from 23 µA to 3 µA at pH 7.0. This indicates that follow-up reactions of the anodically formed oxidation product **II**-$DPAH_2^{+\bullet}$ proceed with higher reaction rate at a solutions pH of 7.0. These processes also can include further anodic oxidation reactions. As an example, further dissociation of **II**-$DPAH_2^{+\bullet}$ to form the radical **III**-$DPAH^{2+\bullet}$, will be more pronounced at pH 7. Thus follow up reactions will be more rapid and the concentration of **II**-$DPAH_2^{+\bullet}$ present in the diffusion layer and available for cathodic reduction during the reverse scan of the voltammogram will be lower at pH 7.0 when compared to pH 5.8.
The overall reaction mechanism of the anodic oxidation of dopamine thus can be formulated in a reaction scheme given in Figure 7. Dopamine is oxidized in an anodic single-electron oxidation (E) to the seminquinone radical, which upon deprotonation (C) stabilizes the remaining proton in between the two phenolic oxygen atoms, which is in agreement with ^{13}C-NMR shifts published elsewhere [25]. The electron transfer is a fast reaction and is independent of pH. The following dissociation reaction establishes a fast pH dependent equilibrium reaction. The overall reversible EC reaction leads to the formation of the semiquinone radical **II**-$DPAH_2^{+\bullet}$. Dimerisation as the next step towards irreversible polydopamine formation will depend on the semiquinone radical concentration **II**-$DPAH_2^{+\bullet}$ which is higher at pH 7.0.

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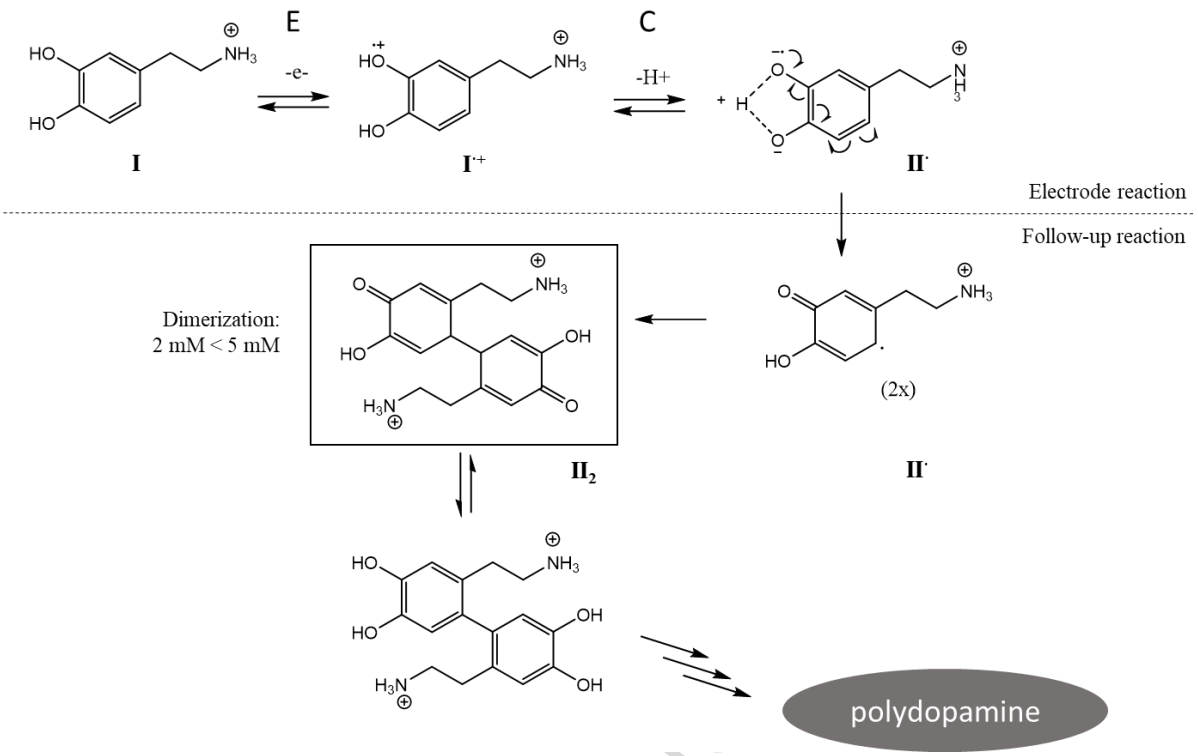


Figure 7: Proposal for the predominating mechanism consisting of an anodic oxidation at the electrode and follow-up reactions in buffered solution at pH 5.8 and 7.0 leading to formation of polydopamine.

The mechanism proposed in Figure 7 is in agreement with reaction steps for polydopamine formation published in the literature [29]. Analytical results obtained with NMR and mass spectroscopy indicated that formation of ring-bridged oligomers will be part of the reaction pathway towards polydopamine. In this study the formation of reactive intermediate products will occur in the diffusion layer of the electrode only. Furthermore the anodic follow-up reactions will make identification or isolation of individual chemical compound difficult.

5. Conclusions

Calculation of the pH dependent species distribution of dopamine indicates that different species have to be considered in the discussion of the anodic oxidation of dopamine.
In the pH range of 5.8 – 7.0 the protonated **I**-$DPAH_3^+$ form of dopamine is the prevailing species. Thus for the anodic oxidation of dopamine an EC mechanism leading to the

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corresponding semiquinone **II**-$DPAH_2^{+\bullet}$ is proposed as the first step of the oxidation on a gold electrode.

The minor dependency of the anodic peak currents $(I_p)_c$ from the solutions pH and the electrochemical reversibility of the electrode process support the presence of a fast EC reaction mechanism.

The radical semiquinone **II**-$DPAH_2^{+\bullet}$ forms the basis for a possible dimerisation leading to catecholamine type structures. At pH 7.0, follow-up reactions e.g. dimerisation of the initially formed semiquinone, consecutive deprotonation, further anodic oxidation reactions proceed at higher rate. Thus lower peak current $(I_p)_c$ for cathodic reduction of the semiquinone was observed at this pH.

The presence of a semiquinone radical as an intermediate in the oxidative polymerization of dopamine explains the formation of catecholamine based oligomers as precursors in polydopamine formation.

The calculation of the species distribution also indicates that the amino group of dopamine will remain protonated in the pH range 5.8 – 7.0 thus involvement of Michael addition and Schiff type reactions with cyclisation to indole ring containing products may occur only with lower reaction rate. These reactions thus will be of less significance for the reactions observed in CV experiments.

At higher pH e.g. pH 8 or pH 10 the presence of different species as starting products for (auto)oxidation or anodic oxidation has to be considered. Also dissociation of oxidation products will depend on the actual pH value and thus the reaction pattern may change.

Due to the high reactivity of dopamine and the complex pattern of reaction pathways, many different mechanisms have been proposed. The present study demonstrates the importance of a pH-dependent species distribution for analysis of reaction mechanisms in oxidation.

In view of the concept of species distribution, it becomes clear that already slight changes in conditions may change the reaction pattern considerably both in terms of reaction rates and with regard to products formed.

Acknowledgement

This research was financially supported by the Austrian research promotion agency (FFG) K-Project tccv (860474) Textile Competence Center Vorarlberg.

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Highlights

- At pH 5.8 – 7.0 an EC reaction is proposed as first step of anodic dopamine oxidation
- pH dependent species distribution of dopamine determines the reaction pathway
- Formation of a semiquinone radical explains formation of ring-bridged polydopamine
- pH dependent follow-up reactions proceed more fast at pH 7.0
- The first step of the anodic dopamine oxidation is electrochemically reversible

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